ETHAN KNOX

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SUMMARY

Computational physicist and applied mathematician completing an M.S. in Applied Mathematics at UIUC (May 2026), with dual B.S. degrees in Mathematics and Physics from Purdue. Research spans nonlinear PDE analysis, GPU-accelerated scientific computing, and biophysical image processing. Proficient in Python, C/C++, CUDA, and the scientific computing stack. Seeking roles in scientific software engineering, data science, or quantitative research.

EDUCATION

University of Illinois — Urbana-Champaign Champaign, IL
M.S. in Applied Mathematics *Aug 2022 – May 2026*

Purdue University West Lafayette, IN
B.S. in Mathematics, Minor in Psychology | GPA 3.5 *Jun 2017 – May 2021*
B.S. in Physics, Minor in Astronomy | GPA 3.5

- Dean's List; Semester Honors; Provost Summer Scholarship; Frank O'Bannon Honors & Accelerated Scholarships.

RESEARCH & EXPERIENCE

University of Illinois — Urbana-Champaign, Kirr Research Group Champaign, IL
Graduate Research Assistant — Dept. of Mathematics *Aug 2025 – Dec 2025*

- Co-authored research report on existence, uniqueness, and stability of multi-peak ground-state solutions of the nonlinear Schrödinger equation, with applications to optical fiber pulse propagation.
- Reduced an infinite-dimensional PDE to a tractable finite-dimensional algebraic system using concentration-compactness and Lyapunov–Schmidt decomposition techniques.
- Classified admissible multi-peak geometric configurations (collinear, isosceles, equilateral, rotationally parameterized) by deriving explicit constraints on Hessian eigenvalue ratios and orientation angles.
- Resolved kernel obstructions in the linearized system by identifying translational and rotational symmetries, enabling continuation of solutions via the implicit function theorem.

Purdue University — Iyer-Biswas Lab West Lafayette, IN
Undergraduate Research Assistant *Aug – Dec 2020*

- Developed custom Python scripts to analyze and process microscopy image data of individual cell growth, contributing to published research on stochastic single-cell behavior (iyerbiswas.com/research).
- Built automated image post-processing pipelines to extract quantitative measurements from large experimental datasets.
- Collaborated cross-functionally with experimentalists and theorists; presented progress and edge-case analysis in weekly all-staff meetings.

The Tap West Lafayette, IN
Bar-Back *Oct 2021 – Jun 2022*

- Operated in a fast-paced, high-throughput environment; demonstrated sustained adaptability and cross-functional teamwork.

PROJECTS

Nulltracer — GPU-Accelerated Black Hole Ray Tracer github.com/ethank5149
CUDA • Python / CuPy • FastAPI • General Relativity • Numerical ODEs • Docker

- Developed a CUDA server application that traces null geodesics through the Kerr-Newman spacetime metric (rotating, charged black hole), rendering photon rings, gravitational lensing, and accretion-disk Doppler effects.
- Implemented five high-order integrators (RK4, Yoshida 4th/6th/8th-order symplectic, RKDP8 adaptive); first-order equation refactor achieved ~40% speedup over the original Hamiltonian formulation.
- Deployed via Docker + NVIDIA Container Toolkit with a thin browser client for real-time interactive parameter exploration; used by the community at v0.9.

SKILLS

Programming: Python (NumPy, SciPy, SymPy, Pandas, Matplotlib, CuPy, FastAPI), C/C++ (Boost, XTensor), LaTeX, Git, Bash / Linux

Data & Analysis: Statistical modeling, image processing, stochastic process analysis, data visualization, scientific computing, CUDA / GPU computing

Mathematics: PDEs, functional analysis, numerical methods, linear algebra, probability & statistics, dynamical systems, optimization, tensor analysis

Physics: General relativity, astrophysics, quantum mechanics, classical mechanics, nonlinear optics, computational physics

ADDITIONAL

Eagle Scout — Boy Scouts of America | **Brazilian Jiu-Jitsu Practitioner** | Dean's List | Semester Honors